

Vectors $u = \langle u_1, u_2, u_3 \rangle$, $v = \langle v_1, v_2, v_3 \rangle$, c is a constant.

$$u + v = \langle u_1 + v_1, u_2 + v_2, u_3 + v_3 \rangle$$

$$u - v = \langle u_1 - v_1, u_2 - v_2, u_3 - v_3 \rangle$$

$$cu = \langle cu_1, cu_2, cu_3 \rangle$$

Dot Product

$$u \cdot v = u_1 v_1 + u_2 v_2 + u_3 v_3.$$

$$|u| = \sqrt{u \cdot u} = \sqrt{u_1^2 + u_2^2 + u_3^2} \quad \leftarrow \text{this is called the length of } u.$$

Calculate the Determinants

① 2x2 matrix

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

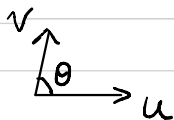
② Higher Dimensional Matrix

$$\det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$$

$$= a(-1)^{1+1} \det \begin{pmatrix} e & f \\ h & i \end{pmatrix} + b(-1)^{1+2} \det \begin{pmatrix} d & f \\ g & i \end{pmatrix}$$

$$+ c(-1)^{1+3} \det \begin{pmatrix} d & e \\ g & h \end{pmatrix}$$

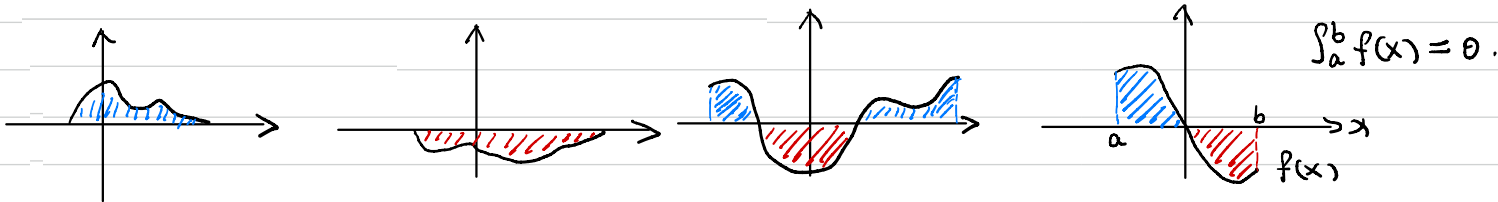
Angle between the Vectors



$$u \cdot v = |u||v| \cos(\theta)$$

Geometrical Meaning of Integrals

— positive
— negative



u-substitution

$$\int_a^b f(x) dx$$

- ① Determine what is u ? ($u = g(x)$)
- ② Find $\frac{du}{dx}$ and derive $dx = ? du$
- ③ Find new lower and upper bounds
 $c = g(a)$ $d = g(b)$
- ④ Compute the new integral about $\int_c^d h(u) du$.

Integration by Parts

$$\int u dv$$

- ① Determine what is u and dv ?
- ② Find du and v .
- ③ Plug u, v, du, dv into $\int u dv = uv - \int v du$.

Find Min or Max of $f(x)$ on an interval $[a, b]$

① Find Critical Points

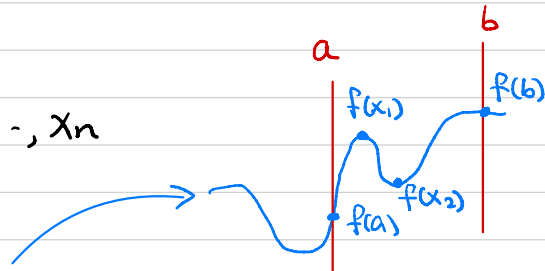
1) Find $f'(x)$

2) Set $f'(x)=0$ and Solve for $x: x_1, \dots, x_n$

② keep only critical points between $[a, b]: x_m, \dots, x_n$

③ Plug in those $x_m, \dots, x_n$ to $f(x)$

④ Find the boundary values $f(a), f(b)$. why?



⑤ The solution is the min or max of $f(a), f(b), f(x_1), \dots, f(x_n)$

Product Rule

$$f(x) = g(x)h(x)$$

$$f'(x) = g(x)h'(x) + g'(x)h(x)$$

Chain Rule

$$f(x) = g(h(x))$$

$$f'(x) = g'(h)h'(x)$$

Right-handed Rule

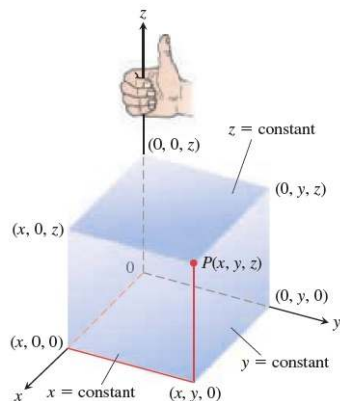
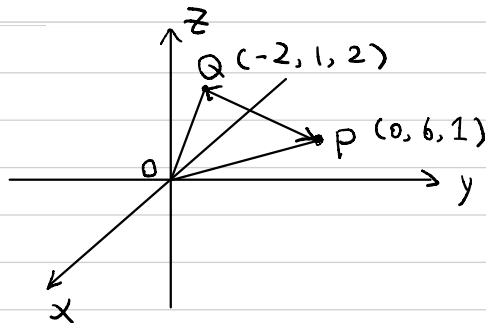


FIGURE 12.1 The Cartesian coordinate system is right-handed.

Vectors in 3-D space



$$\vec{PQ} = \vec{OQ} - \vec{OP}$$

$$= \langle -2, 1, 2 \rangle - \langle 0, 6, 1 \rangle$$

$$= \langle -2, -5, 1 \rangle$$

$$\vec{QP} = \vec{OP} - \vec{OQ}$$

$$= \langle 0, 6, 1 \rangle - \langle -2, 1, 2 \rangle$$

$$= \langle 2, 5, -1 \rangle$$

Cross Products

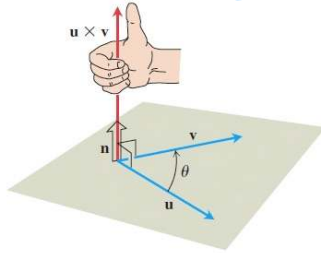


FIGURE 12.28 The construction of $\mathbf{u} \times \mathbf{v}$.

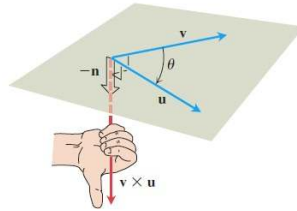


FIGURE 12.29 The construction of $\mathbf{v} \times \mathbf{u}$.

Def: $\mathbf{u} \times \mathbf{v} = |\mathbf{u}| |\mathbf{v}| (\sin \theta) \mathbf{n}$ ↖ unit vector in the orthogonal direction to both $\mathbf{u}$ and $\mathbf{v}$.

- The cross product $\mathbf{u} \times \mathbf{v}$ is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$
i.e. $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{u} = 0$ and $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{v} = 0$
- Nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ are parallel if and only if $\mathbf{u} \times \mathbf{v} = \mathbf{0}$
- $|\mathbf{u} \times \mathbf{v}|$ is area of a parallelogram formed by $\mathbf{u}$ and $\mathbf{v}$

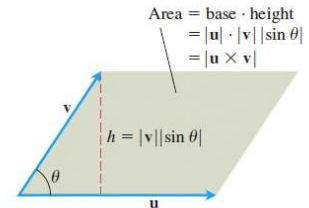


FIGURE 12.31 The parallelogram determined by $\mathbf{u}$ and $\mathbf{v}$.

How to compute the cross product?

$$u = \langle u_1, u_2, u_3 \rangle, \quad v = \langle v_1, v_2, v_3 \rangle$$

$$\begin{aligned} u \times v &= \begin{vmatrix} i & j & k \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = i(-1)^{+1} \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} + j(-1)^{+2} \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} + k(-1)^{+3} \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \\ &= (u_2 v_3 - u_3 v_2) i - (u_1 v_3 - u_3 v_1) j + (u_1 v_2 - u_2 v_1) k \\ &= \langle u_2 v_3 - u_3 v_2, -(u_1 v_3 - u_3 v_1), u_1 v_2 - u_2 v_1 \rangle \end{aligned}$$

Rickey's OHs: 3-4 pm Mondays @ Clough 280 (Mathlab Drop-in Tutor).

Expression for a line

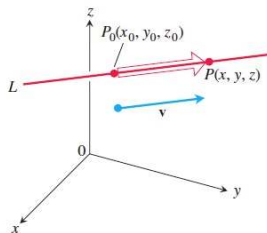


FIGURE 12.36 A point P lies on L through P_0 parallel to $\mathbf{v}$ if and only if $\vec{P_0P}$ is a scalar multiple of $\mathbf{v}$.

Determined by a point and a direction.

$P_0: (x_0, y_0, z_0)$ vector $\vec{v} = \langle v_1, v_2, v_3 \rangle$

The expression of the line through P_0 in the direction of $\vec{v}$ is

$$\begin{aligned} \mathbf{r}(t) &= \langle x_0, y_0, z_0 \rangle + t \langle v_1, v_2, v_3 \rangle \\ &= \langle x_0 + tv_1, y_0 + tv_2, z_0 + tv_3 \rangle \end{aligned}$$

Equation for a plane

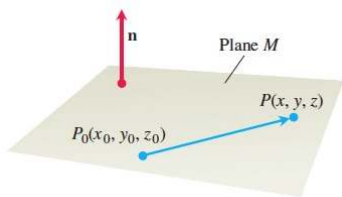


FIGURE 12.40 The standard equation for a plane in space is defined in terms of a vector normal to the plane: A point P lies in the plane through P_0 normal to $\mathbf{n}$ if and only if $\mathbf{n} \cdot \vec{P_0P} = 0$.

Determined by a point and a normal vector

$P_0: (x_0, y_0, z_0)$ $\vec{n} = \langle A, B, C \rangle$

vector equation: $\mathbf{n} \cdot \vec{P_0P} = 0$

Component equation: $A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$

Component equation simplified: $Ax + By + Cz = Ax_0 + By_0 + Cz_0$.

Find the line of intersection of two planes $A_1x + B_1y + C_1z = D_1$
and $A_2x + B_2y + C_2z = D_2$

- Direction → ① Find two normal vectors: $\langle A_1, B_1, C_1 \rangle$ & $\langle A_2, B_2, C_2 \rangle$
Base Point → ② Find the cross product $\langle A_1, B_1, C_1 \rangle \times \langle A_2, B_2, C_2 \rangle := \vec{v}$
③ Find a point on the line of intersection: (x_0, y_0, z_0)
one method: set $z=0$, then solve for

$$\begin{cases} A_1x + B_1y = D_1 \\ A_2x + B_2y = D_2 \end{cases}$$

- ④ Write out the expression of the line: $\langle x_0, y_0, z_0 \rangle + t\vec{v}$

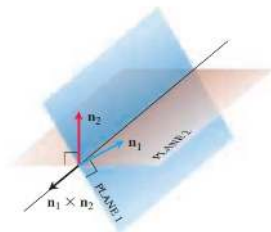


FIGURE 12.41 How the line of intersection of two planes is related to the planes' normal vectors (Example 8).

Find a plane through 3 points: $P(p_1, p_2, p_3), Q(q_1, q_2, q_3), R(r_1, r_2, r_3)$.

① Check the three lines are not collinear.
if collinear, find an arbitrary point off this line.

② Find two vectors, say $\vec{PQ}, \vec{PR}$

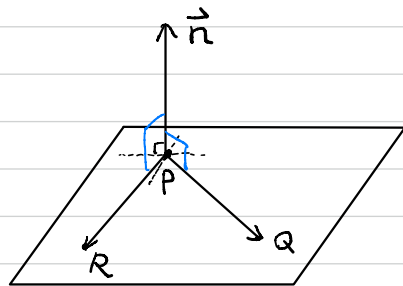
normal vector $\rightarrow$ ③ Use cross products to find the normal vector of the plane $\vec{n} := \vec{PQ} \times \vec{PR}$

a point $\rightarrow$ ④ Use any point on the plane (for example any of P, Q , or R) to write out the equation of the plane:

If $\vec{n} = \langle A, B, C \rangle$, then the plane is

$$A(x - p_1) + B(y - p_2) + C(z - p_3) = 0$$

$$\Rightarrow Ax + By + Cz = Ap_1 + Bp_2 + Cp_3.$$



Some Common 3D objects:

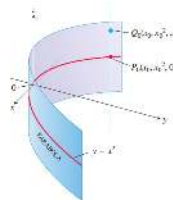
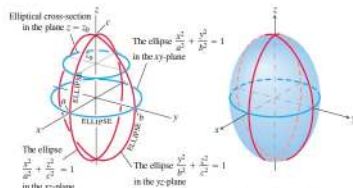
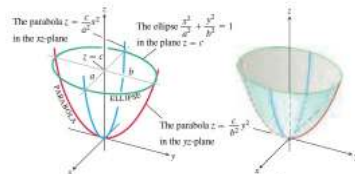


FIGURE 12.45 Every point of the cylinder in Example 1 has coordinates of the form (x_1, y_1^2, z_1) . We call it "the cylinder $y = x^2$ ".



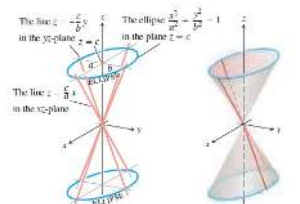
ELLIPSOID

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$



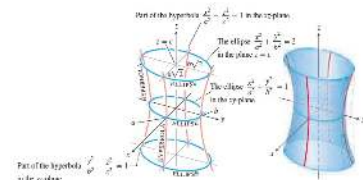
ELLIPTICAL PARABOLOID

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z}{c}$$



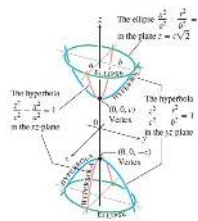
ELLIPTICAL CONE

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}$$



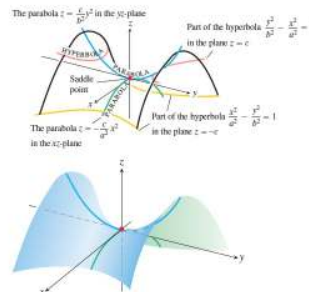
HYPERBOLOID OF ONE SHEET

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$



HYPERBOLOID OF TWO SHEETS

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$



HYPERBOLIC PARABOLOID

$$\frac{y^2}{b^2} - \frac{x^2}{a^2} = \frac{z}{c}, c > 0$$

Parametric Egn of a Curve

$$\mathbf{r}(t) = \langle x(t), y(t) \rangle$$

This is also the position vector

- To find the path given a parametric eqn, that is find $y(x)$
Don't forget the domain, which is what value $x(t)$ takes on.

Position, Velocity, Acceleration

- Position vector: $\mathbf{r}(t) = \langle x(t), y(t) \rangle$
- Velocity vector: $\mathbf{r}'(t) = \langle x'(t), y'(t) \rangle$
- Acceleration vector: $\mathbf{r}''(t) = \langle x''(t), y''(t) \rangle$

Parametric Egn of Tangent Line

Given $\mathbf{r}(t) = \langle x(t), y(t) \rangle$

$$\Rightarrow \mathbf{r}'(t) = \langle x'(t), y'(t) \rangle$$

- The Tangent line at $t = t_0$ is

$$\mathbf{S}(k) = \mathbf{r}(t_0) + k \mathbf{r}'(t_0) = \langle x(t_0), y(t_0) \rangle + k \langle x'(t_0), y'(t_0) \rangle.$$

Take Integrals

If $r''(t) = \langle a(t), b(t) \rangle$, w/ initial condition $r'(t_0) = \vec{d}$, $r(t_0) = \vec{e}$

Then we have

$$r'(t) = \int \langle a(t), b(t) \rangle dt = \langle \int a(t) dt, \int b(t) dt \rangle + \vec{C}_1 \quad \leftarrow \text{a vector.}$$

Then plug in $r'(t_0) = \vec{d}$, we have

$$\langle \int a(t) dt, \int b(t) dt \rangle + \vec{C}_1 = \vec{d}$$
$$\Rightarrow \vec{C}_1 = \vec{d} - \langle \int a(t) dt, \int b(t) dt \rangle$$

$$\text{Let } r'(t) = \langle f(t), g(t) \rangle = \langle \int a(t) dt, \int b(t) dt \rangle + \vec{C}_1$$

Then we have

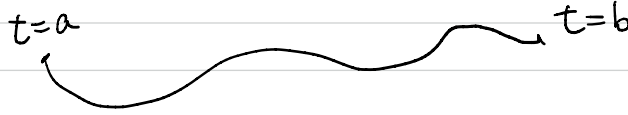
$$r(t) = \int \langle f(t), g(t) \rangle dt = \langle \int f(t) dt, \int g(t) dt \rangle + \vec{C}_2 \quad \leftarrow \text{a vector.}$$

Then plug in $r(t_0) = \vec{e}$, we have

$$\langle \int f(t) dt, \int g(t) dt \rangle + \vec{C}_2 = \vec{e}$$
$$\Rightarrow \vec{C}_2 = \vec{e} - \langle \int f(t) dt, \int g(t) dt \rangle$$

If there is a domain for $r''(t)$, make sure to keep the domain in the solution $r(t)$ and $r'(t)$

Arc Length

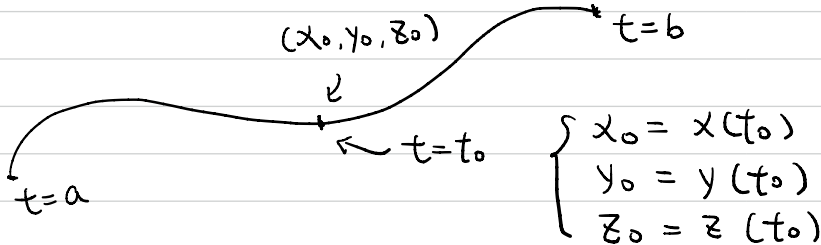


$$\mathbf{r}(t)$$
$$\mathbf{v}(t) = \mathbf{r}'(t) \text{ velocity}$$

$$L = \int_a^b |\mathbf{v}(t)| dt = \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2} dt.$$

Correspondence between points and parameter.

$$\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$$



Unit Tangent Vector (T)

A function about t that describes the unit tangent line of a parametrized curve $\mathbf{r}(t)$ at time t .

$$\mathbf{T}(t) = \frac{\mathbf{v}(t)}{|\mathbf{v}(t)|}$$

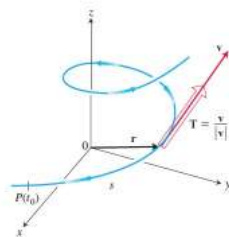


FIGURE 13.15 We find the unit tangent vector $\mathbf{T}$ by dividing $\mathbf{v}$ by its length $|\mathbf{v}|$.

Curvature (K)

The rate at which $\mathbf{T}$ turns per unit of length along the curve

$$K = \frac{1}{|\mathbf{v}(t)|} \left| \frac{d\mathbf{T}}{dt} \right|$$

Principal Unit Normal (N)

$\frac{d\mathbf{T}}{ds}$ divided by its length K , orthogonal to $\mathbf{T}$.

$$\mathbf{N} = \frac{1}{K} \frac{d\mathbf{T}}{ds} = \frac{d\mathbf{T}/dt}{|d\mathbf{T}/dt|}$$

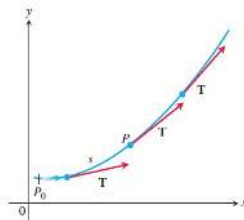


FIGURE 13.17 As P moves along the curve in the direction of increasing arc length, the unit tangent vector turns. The value of $|d\mathbf{T}/ds|$ at P is called the *curvature* of the curve at P .

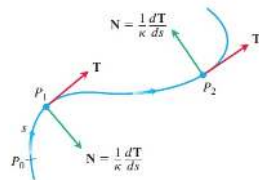


FIGURE 13.19 The vector $d\mathbf{T}/ds$, normal to the curve, always points in the direction in which $\mathbf{T}$ is turning. The unit normal vector $\mathbf{N}$ is the direction of $d\mathbf{T}/ds$.